A Bayesian Joint Longitudinal–Interval and Multi-State Threshold-Crossing Framework for Irregular Molecular Monitoring in Chronic Myeloid Leukemia

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Abstract

Routine molecular monitoring in chronic myeloid leukemia (CML) produces data that violate standard survival and landmark assumptions: log-transformed minimal residual disease (log-MRD) is measured at irregular visits, is left-censored at an assay reporting floor, and deep molecular response (DMR) is detected only when tested. We develop a Bayesian framework with two linked models. The first couples a left-censored longitudinal log-MRD sub-model with an interval-observed likelihood for first documented DMR through shared random effects, fitted by Hamiltonian Monte Carlo. The second reads the endpoint as a latent threshold-crossing (first-hitting-time) process and extends it to a multi-state model in which DMR onset, durable response, and molecular relapse are crossings of one latent trajectory; because the trajectory is piecewise-quadratic, its running minimum and maximum are closed-form, giving divergence-free sampling. In a Hamiltonian Monte Carlo simulation, the variance components and the biomarker–event association attain near-nominal credible-interval coverage, while the latent time trend is attenuated under heavy assay-floor censoring. In an 87-patient cohort the joint model was calibrated in the large and improved on simpler alternatives by Brier score, while the multi-state model captured onset and, with a spline trajectory, relapse. Dynamic predictions are reported as internal accuracy.

Keywords: joint longitudinal–survival models; interval-censored events; left-censored biomarkers; Bayesian hierarchical modelling; dynamic prediction; chronic myeloid leukemia; deep molecular response.

1 Introduction

Tyrosine kinase inhibitor (TKI) therapy has transformed the prognosis of chronic myeloid leukemia (CML), shifting long-term care toward sustained molecular control, adherence, adverse-effect management, and the possible selection of patients for treatment-free remission [1–4]. Deep and durable molecular response, including MR4.5 and deep molecular response (DMR), is central to long-term response assessment and to discussions about treatment discontinuation [5–8]. Large real-world cohorts confirm that the depth, timing and durability of molecular response drive long-term outcome: early molecular response (BCR-ABL $\leq 10\%$ at three months) predicts superior progression-free and overall survival by landmark analysis [9, 10], loss of an achieved response counts as an event, and outcomes worsen the later response is attained [11]. Consequently, statistical models for CML are increasingly asked to turn longitudinal molecular histories into individualized, calibrated statements about response.

Routine molecular monitoring data, however, rarely have the structure assumed by simple survival or prediction models. Three features recur. First, measurements are obtained at *irregular* clinical visits, so the exact time at which a patient first attains DMR is never observed; only the interval between two visits that brackets the transition is known. Treating first *observed* DMR as an exact event time ignores this interval observation and confounds biological onset with visit timing. Second, deep molecular values are frequently reported at an assay or reporting floor. Treating floor values as exact numerical measurements understates uncertainty below the reporting limit; the correct information is that the true burden lies at or below the floor. Third, and more subtly, the clinical endpoint is *defined* by a threshold on the same biomarker that any joint model uses as a predictor: DMR is log-MRD ≤ -4.5. Using the latent biomarker to predict a separately coded DMR "event" can therefore be circular.

Joint longitudinal–survival models provide a natural framework for connecting repeated biomarkers with event processes [12, 13], and recent methodological work has extended them to complex observation schemes and flexible biomarker–event associations [14, 15]. What is missing for molecular monitoring is a single framework that simultaneously (i) handles assay-floor left-censoring in the biomarker, (ii) respects the interval nature of endpoint detection under irregular visits, and (iii) links the event to the latent biomarker trajectory—with a threshold-crossing formulation that removes the residual definitional circularity between predictor and endpoint developed as an extension—while (iv) delivering calibrated, individualized dynamic predictions without overstating clinical decision utility.

Our event model belongs to the *first-hitting-time* (first-passage-time) tradition, in which failure is the first time a latent stochastic process reaches a boundary: threshold regression [16] and the process point of view on hazards [17, 18]. What is new here is to bring this tradition to *interval-observed, assay-floor left-censored biomarker monitoring* of first DMR, combining interval-censoring methodology [19], landmarking and joint-model dynamic prediction [20, 21], and proper scoring rules for calibration assessment [22].

This paper develops and illustrates, on a real-world CML cohort, a Bayesian framework built around two linked models: a left-censored joint longitudinal–interval model for the time to first documented DMR, and a latent threshold-crossing and multi-state model that reads molecular response directly off the latent trajectory. Its contributions are six. First, a longitudinal sub-model for the latent log-MRD trajectory that treats assay-floor measurements as left-censored rather than as exact values (Section 3.1). Second, an interval-observed likelihood for first documented DMR that respects the fact that response is ascertained only at monitoring visits (Section 3.2). Third, a latent threshold-crossing formulation that removes the definitional circularity between the biomarker predictor and the DMR endpoint, extended to a *multi-state* process—onset, durable response, and molecular relapse—as successive crossings of one latent trajectory, with a piecewise-quadratic trajectory whose running minimum and maximum are available in closed form and therefore sample without divergences (Section 3.3). Fourth, fully Bayesian estimation of both models by Hamiltonian Monte Carlo. Fifth, a simulation study—structured according to the ADEMP framework, with a comprehensive Hamiltonian Monte Carlo evaluation ongoing—whose preliminary results indicate that separating the threshold width from measurement error improves its identifiability and that coding assay-floor values as exact attenuates between-patient heterogeneity (Section 5). Sixth, an application to the cohort with posterior predictive checks, calibration, model comparison, dynamic prediction reported as internal (apparent) accuracy, and a multi-state fit that captures onset and, with a spline trajectory, molecular relapse (Section 4). Joint modelling of the informative visit process is outlined as future work; external validation is the subject of ongoing work.

2 Motivating application: a CML molecular-monitoring cohort

We motivate and illustrate the methodology with a real-world CML monitoring cohort assembled from routine tyrosine kinase inhibitor monitoring at a single haematology centre in Kunming, Yunnan Province, China. Monitoring followed the 2011 Chinese CML guideline [23], which prescribes response-adaptive molecular testing—quarterly BCR-ABL quantification, intensified to every one-to-three months if the transcript rises—so visit times are irregular and partly response-driven by design. The raw visit-level file contained 504 molecular records on 89 patients. After removing records with missing or invalid essential modelling fields, the longitudinal analysis file contained 495 complete molecular observations from 87 patients; the corresponding patient-level file contained 87 records and 275 at-risk intervals for first documented DMR (Table 1). Analysis files used coded identifiers; direct identifiers were held only in a private linkage file.

Table 1: Data cleaning audit and generated analysis datasets.

Cleaning step	Count
raw_longitudinal_rows	504
raw_longitudinal_patients	89
dropped_missing_or_invalid_core_fields	9
model_ready_longitudinal_rows	495
model_ready_longitudinal_patients	87
raw_patient_table_rows	87
usable_patient_covariate_rows	71
patient_level_analysis_rows	87
complete_core_covariate_rows	62
interval_survival_rows	275
dmr_event_patients	68
censored_patients	19

DMR was defined as log-MRD ≤ -4.5 and complete molecular response (CMR) as log-MRD ≤ -5.0; both indicators were audited by recomputation from the processed values. A structural feature of the data is central to the modelling: 239 of 495 observations (48.3%) were at or below the log-MRD floor of -5.0, and no retained observation fell in the interval $(-5.0, -4.5]$. As a result, DMR-positive and CMR-positive counts coincide in the retained data — not through a coding error, but because all DMR-positive values were reported at the −5.0 assay floor. This directly motivates left-censored modelling of floor observations and limits the ability of the data alone to distinguish DMR from deeper response. Treatment time was recorded in months for clinical summaries and converted to years for all likelihood components; if $t^{(m)}$ denotes months since imatinib initiation, the model uses $t = t^{(m)}/12$, with time effects on the $\log(1+t)$ scale.

Monitoring was heterogeneous, with median follow-up of 39.0 months and a median of 5 visits per patient. The median visit gap was 6.0 months; 50.3% of gaps exceeded 6 months and 20.0% exceeded 12 months. This irregularity, together with the fact that DMR is documented only at visits, is precisely what the interval and threshold-crossing likelihoods below are designed to accommodate.

3 Statistical framework

3.1 Longitudinal sub-model with assay-floor left-censoring

Let $m_i(t)$ denote the latent biological log-MRD trajectory for patient i at treatment time t (years):

$$m_i(t) = \beta_0 + \beta_1 \log(1+t) + \beta_2\{\log(1+t)\}^2 + b_{0i} + b_{1i}\log(1+t),$$

with independent patient-specific random effects

$$b_{0i} \sim N(0, \tau_{b0}^2), \qquad b_{1i} \sim N(0, \tau_{b1}^2).$$

An earlier specification included a random intercept–slope correlation; because it was weakly identified in this cohort, the final model treats the random effects as independent, which improved computational stability without materially changing the fit. The observation-scale mean includes a sample-source adjustment for bone-marrow versus peripheral-blood assays:

$$\mu_{ij} = m_i(t_{ij}) + \beta_{\mathrm{BM}}\, I(\text{bone marrow}_{ij}).$$

For observations above the reporting floor, $y_{ij} \sim N(\mu_{ij}, \sigma_y^2)$. For observations reported at the assay floor $c_F = -5.0$, the likelihood contribution is left-censored, carrying the information that the true value lies at or below the floor:

$$\Pr(y_{ij}^* \le c_F) = \Phi\left\{\frac{c_F - \mu_{ij}}{\sigma_y}\right\}.$$

3.2 Interval-observed sub-model for first documented DMR

The event sub-model is an interval-observed likelihood for the time of first *documented* DMR. Because DMR is detected only at monitoring visits, the event is interval-observed. For each patient the first at-risk interval ending in a DMR-positive measurement is coded as the event interval, and preceding intervals as at risk without event; patients without documented DMR by last follow-up are right-censored. Let L_{ik} and R_{ik} be the start and end of at-risk interval k, with $\Delta_{ik} = R_{ik} - L_{ik}$ and $t_{ik}^{mid} = (L_{ik} + R_{ik})/2$. The interval hazard links the latent trajectory to DMR:

$$h_{ik}^{\mathrm{DMR}} = \exp\{\gamma_0 + \gamma_1 \log(1+t_{ik}^{mid}) + \gamma_2 \log(1+\Delta_{ik}) + \alpha\, m_i(t_{ik}^{mid})\}, \qquad p_{ik}^{\mathrm{DMR}} = 1 - \exp\{-h_{ik}^{\mathrm{DMR}}\Delta_{ik}\}.$$

The interval length Δ_{ik} enters in two distinct roles that should not be conflated. It first enters as *exposure time* through the cumulative-hazard form $p_{ik}^{\mathrm{DMR}} = 1 - \exp\{-h_{ik}^{\mathrm{DMR}}\Delta_{ik}\}$, the standard complementary log–log likelihood for a grouped or interval-censored event; this encodes the baseline assumption that the opportunity to document DMR accumulates *proportionally* with time at risk. The additional term $\gamma_2 \log(1 + \Delta_{ik})$ sits inside the log-hazard and does not re-express interval length as exposure: it modulates the per-unit-time documentation *intensity* and therefore measures *departure from proportional exposure accumulation.* A negative γ_2 indicates that longer at-risk gaps carry a lower documentation intensity per unit time than proportional accumulation would predict—an observation- and monitoring-process feature (for example, sparser or less informative sampling across long gaps), not a biological property of molecular response. Because the two contributions enter multiplicatively and in different roles—exposure duration versus a log-linear modifier of intensity—the specification does not double-count interval length. The shared latent value $m_i(t_{ik}^{mid})$ couples the two sub-models, so that lower latent MRD raises the probability of documented DMR; this coupling carries the biological interpretation. To confirm that the substantive association is not an artefact of the observation-process term, we refit the event sub-model without $\gamma_2 \log(1 + \Delta_{ik})$ as a sensitivity analysis (reported in Section 4).

3.3 Latent threshold-crossing and multi-state model

The interval sub-model treats first DMR as an interval-censored event whose likelihood is driven by the biomarker, but DMR is *defined* by the latent log-MRD trajectory crossing $c_D = -4.5$. Our second model represents this directly as a first-hitting-time process [16, 17], which removes the definitional circularity between the biomarker predictor and the endpoint. Writing $M_i(t) = \min_{0 \le u \le t} m_i(u)$ for the running minimum of the latent trajectory, first DMR is $T_i^D = \inf\{t \ge 0 : m_i(t) \le c_D\} = \inf\{t : M_i(t) \le c_D\}$, so a patient with documented first DMR in $(L_i, R_i]$ satisfies $M_i(L_i) > c_D$ and $M_i(R_i) \le c_D$, and a censored patient satisfies $M_i(C_i) > c_D$. For gradient-based sampling we replace the hard indicators by a probabilistic crossing with an effective patient-level threshold $C_i \sim N(c_D, \sigma_{\text{thr}}^2)$: an event documented in $(L_i, R_i]$ contributes $\Phi\{(M_i(L_i) - c_D)/\sigma_{\text{thr}}\} - \Phi\{(M_i(R_i) - c_D)/\sigma_{\text{thr}}\}$ and a censored patient the corresponding tail. Separating the threshold width σ_{thr} (threshold ambiguity) from the measurement error σ_y makes both identifiable, whereas a single width confounds them. In this cohort every documented DMR was reported at the -5.0 floor, so the observed endpoint is operationally first attainment of the floor and σ_{thr} additionally absorbs the gap between the nominal -4.5 cut-point and the recorded value.

Multi-state extension. Molecular response has depth and is reversible: a patient may attain DMR, sustain it, or lose it, and treatment-free-remission eligibility depends on *durable* response [9, 11]. We therefore extend the crossing model to a bidirectional multi-state process on the same latent trajectory, with states 0 (not yet DMR), 1 (in DMR) and 2 (molecular relapse). Onset ($0 \to 1$) is the downward crossing above; durable response over a window w is the event $\max_{u \in [T_i^D, T_i^D + w]} m_i(u) \le c_D$ (the running *maximum* after onset staying below threshold); and relapse ($1 \to 2$) is the first upward crossing $T_i^R = \inf\{t > T_i^D : m_i(t) > c_D\}$. Each transition enters the likelihood through the same probit crossing device with width σ_{thr}. Because m_i is quadratic in $\log(1 + t)$, both extrema are available in *closed form*—the running minimum is the value at the vertex $\ell^* = -(\beta_1 + b_{1i})/(2\beta_2)$ clamped to the observed range, and past the vertex m_i is monotone, so the upward crossing and the durability maximum are point evaluations. This removes the soft-minimum grid approximation entirely and yields a smooth log density that samples without the divergent transitions the grid induces.

To capture individual molecular relapse the trajectory itself is made flexible while retaining the closed form: adding a patient-specific slope change $\zeta_i \sim N(0, \sigma_\zeta^2)$ after a knot κ, $m_i(\ell) + = \zeta_i(\ell - \kappa)_+$, makes the trajectory piecewise-quadratic, so patients may rebound after κ and thereby relapse while onset and post-knot values remain analytic. The knot κ is *fixed a priori* at two years ($\kappa = 24$ months) rather than estimated, on clinical grounds: in TKI-treated CML the initial molecular decline is typically achieved and consolidated within the first two years, so a rebound permitted after this point coincides with the clinically relevant window for loss of response. Fixing κ also preserves the computational advantages of the formulation. Estimating κ as a free parameter would make the location of the trajectory's vertex, and hence the closed-form running minimum and maximum, depend on a quantity that is itself being inferred; with only a handful of confirmed relapses the knot is weakly identified and the likelihood becomes multimodal—distinct knot positions explain the sparse post-onset data almost equally well—inducing label-switching between the pre- and post-knot slope regimes and destroying the smooth, divergence-free geometry that the closed-form extrema provide. Sensitivity to the knot location is instead examined by refitting at neighbouring values (18 and 30 months), which leaves the longitudinal and onset estimates essentially unchanged. An optional Gaussian frailty $\rho_i \sim N(0, \sigma_\rho^2)$ on the relapse threshold is equivalent to a wider relapse-specific width $\sigma_{\text{rel}} = \sqrt{\sigma_{\text{thr}}^2 + \sigma_\rho^2}$ and preserves the closed form (Supplementary Material).

3.4 Dynamic prediction and recalibration

For clinical use the framework produces individualized dynamic predictions by landmarking [20, 21]. At landmark time s the target is

$$\Pr(T_i^D \le s + H \mid T_i^D > s, \mathcal{H}_i(s)),$$

where $\mathcal{H}_i(s)$ is the MRD history available up to s, with landmarks $s \in \{6, 12, 18, 24\}$ months and horizons $H \in \{6, 12, 24\}$ months. Because absolute probabilities from a mechanistic joint model can be miscalibrated, we apply horizon-specific logistic recalibration,

$$\operatorname{logit}\{p_{i,\text{cal}}(s, H)\} = a_H + b_H \operatorname{logit}\{p_{i,\text{orig}}(s, H)\},$$

with the recalibration layer assessed by patient-clustered bootstrap optimism correction, since each patient contributes several landmark–horizon records. Strict prospective prediction requires that patient-specific random effects be updated using only $\mathcal{H}_i(s)$; the internal analysis in Section 4 conditions on the full longitudinal record and is therefore an apparent (upper-bound) assessment of calibration, which we treat accordingly.

3.5 Priors, estimation, and computation

Estimation is fully Bayesian via Hamiltonian Monte Carlo in Stan. Weakly informative priors were fixed before fitting:

$$\beta_0 \sim N(-2.5, 2^2), \quad \beta_1 \sim N(-1, 1^2), \quad \beta_2 \sim N(0, 0.5^2), \quad \beta_{\text{BM}} \sim N(0, 1^2),$$

$$\sigma_y \sim \text{Exponential}(1), \quad \tau_{b0}, \tau_{b1} \sim \text{Exponential}(1), \quad \gamma_0 \sim N(-2, 2^2), \quad \gamma_1, \gamma_2 \sim N(0, 1^2), \quad \alpha \sim N(-0.5, 0.75^2).$$

The Exponential(1) priors on the scale parameters $\sigma_y, \tau_{b0}, \tau_{b1}$ are weakly informative but place their mass below the fitted values; because every model compared here uses identical priors, prior choice does not confound the floor-censored versus exact-floor contrast, though a prior-sensitivity analysis (for example half-normal or half-Cauchy scales) is advisable when the absolute magnitude of between-patient heterogeneity is of primary interest. Convergence was assessed by $\widehat{R}$, bulk and tail effective sample size, divergent-transition counts, tree depth, and E-BFMI. In the multi-state threshold-crossing model the running minimum and maximum are evaluated in closed form from the piecewise-quadratic trajectory, so no grid or soft-minimum approximation is needed and the log density is smooth; the probabilistic crossing device (width σ_{thr}) preserves differentiability of the interval-transition terms.

3.6 Model assessment and comparison

Model adequacy was evaluated with posterior predictive checks for the longitudinal component (predictive coverage; observed versus predicted assay-floor rate) and with calibration of DMR probabilities at both the interval and patient level (calibration-in-the-large, calibration slope, and grouped calibration plots). Discrimination and overall accuracy were summarized by the Brier score, a strictly proper scoring rule [22]. The joint model was compared against four simpler alternatives that a practitioner might otherwise use: a descriptive Kaplan–Meier analysis of first observed DMR; an interval-only model with timing and visit-gap terms; a landmark MRD model; and a joint model that treats floor observations as exact -5.0. The Kaplan–Meier comparator yields a marginal survival curve rather than covariate-specific risks; its interval- and patient-level probabilities were obtained by evaluating the fitted curve at each unit's follow-up, and its calibration is correspondingly limited. Refittable simpler models were internally validated by patient-cluster bootstrap with optimism correction; for the Bayesian joint models, fixed-posterior bootstrap summaries are reported.

3.7 Simulation study design

We evaluate the framework with a simulation study structured according to the ADEMP framework (aims, data-generating mechanisms, estimands, methods, performance measures) [22]. The *aim* is to assess bias, interval coverage and width, calibration, and computational reliability of the proposed estimators, under both correctly specified and misspecified conditions, and to compare them with established alternatives.

Data-generating mechanisms. Beyond the baseline that draws truth from the fitted model, the mechanisms cross: (i) *trajectory specification*—correctly specified (piecewise-quadratic) versus misspecified (a smoother monotone decline, and a sharper exponential approach) so that the working quadratic is wrong; (ii) *threshold* placed at both $c_D = -4.5$ and $c_D = -5.0$; (iii) *assay floor*—a single fixed floor versus patient- or assay-dependent floors (heterogeneous censoring); (iv) *measurement error and random effects*—Gaussian versus heavy-tailed (t_3) errors and random effects; (v) *visit process*—non-informative versus informative monitoring of increasing strength; (vi) *relapse frequency*—low, moderate, and high; and (vii) *follow-up*—dense versus sparse late follow-up. Sample sizes are $n \in \{87, 150, 300\}$.

Estimands and methods. The estimands are the longitudinal and event-submodel parameters, the transition probabilities, and horizon DMR probabilities. The principal scenarios are fitted with the *same Hamiltonian Monte Carlo* procedure used in the application, so that posterior bias, Bayesian credible-interval coverage, and computational behaviour of the reported estimator are assessed directly rather than through a marginal-MAP proxy; for selected reduced-size settings we additionally run simulation-based calibration (SBC) to verify the posterior. Competing methods include a standard shared-parameter joint model and an interval-censoring survival model.

Performance measures. We report, separately, bias, posterior interval coverage, interval width, Brier score, calibration slope, and the frequency of computational failures (divergent transitions, low E-BFMI, non-convergence). At least 200 replicates are used for the central scenarios so that the Monte Carlo standard error of a coverage estimate near 0.95 is about 0.015; Monte Carlo standard errors are reported for every summary.

The results reported in Section 5 are a *preliminary* subset obtained with marginal maximum-*a-posteriori* estimation and Wald intervals at reduced replicate counts; they are used to establish feasibility and the qualitative effect of the assay floor. The full Hamiltonian Monte Carlo study described above, including the misspecification and informative-monitoring grids and SBC, is ongoing and will replace them.

4 Application to the CML cohort

We fit both models to the cohort by Hamiltonian Monte Carlo. We first report the joint longitudinal–interval model—posterior estimates, diagnostics, posterior predictive checks, calibration, comparison against simpler alternatives, and dynamic prediction—and then the multi-state threshold-crossing model, which additionally resolves onset, durability, and molecular relapse (Section 4.7).

4.1 Cohort and endpoint summary

The final cohort comprised 87 patients, 495 longitudinal log-MRD observations, and 275 at-risk intervals; 68 patients had documented DMR and 19 were censored. Baseline clinical covariates were complete for 62 patients (71.3%) (Table 2). Molecular response deepened over follow-up, and the assay-floor share rose with time (Table 3), consistent with the trajectory and floor-censoring structure in Figure 1.

Table 2: Baseline characteristics and follow-up summaries.

Characteristic	All	DMR	Censored
Patients, n	87	68	19
Age, mean (SD)	44.6 (15.5); n=62	44.9 (15.5); n=55	42.7 (16.2); n=7
Male sex, n (%)	38 (61.3%); n=62	35 (63.6%); n=55	3 (42.9%); n=7
Imatinib duration, years, median [IQR]	5.0 [2.1, 7.5]; n=62	5.0 [3.2, 7.8]; n=55	2.0 [1.5, 3.8]; n=7
Baseline PH+ %, median [IQR]	100.0 [100.0, 100.0]; n=62	100.0 [100.0, 100.0]; n=55	100.0 [100.0, 100.0]; n=7
3-month PH+ %, median [IQR]	0.0 [0.0, 10.1]; n=44	0.0 [0.0, 10.4]; n=38	0.0 [0.0, 3.8]; n=6
6-month PH+ %, median [IQR]	0.0 [0.0, 0.0]; n=52	0.0 [0.0, 0.0]; n=45	0.0 [0.0, 7.5]; n=7
12-month PH+ %, median [IQR]	0.0 [0.0, 0.0]; n=59	0.0 [0.0, 0.0]; n=52	0.0 [0.0, 0.0]; n=7
Follow-up, months, median [IQR]	39.0 [15.8, 65.8]; n=87	46.5 [23.1, 71.7]; n=68	13.0 [10.8, 24.1]; n=19
Visits per patient, median [IQR]	5 [4, 7]; n=87	6 [4, 8]; n=68	4 [3, 4]; n=19
DMR observed, n (%)	68 (78.2%); n=87	68 (100.0%); n=68	0 (0.0%); n=19

Table 3: Longitudinal molecular monitoring summaries by follow-up window.

Window_months	Observations	Patients	BM_samples	LOG_MRD	DMR	CMR
0-3	67	55	67 (100.0%); n=67	-1.2 [-1.6, -0.8]; n=67	2 (3.0%); n=67	2 (3.0%); n=67
>3-6	36	34	36 (100.0%); n=36	-2.0 [-3.5, -1.1]; n=36	8 (22.2%); n=36	8 (22.2%); n=36
>6-12	79	59	79 (100.0%); n=79	-2.7 [-5.0, -1.6]; n=79	26 (32.9%); n=79	26 (32.9%); n=79
>12-24	95	59	92 (96.8%); n=95	-5.0 [-5.0, -2.4]; n=95	51 (53.7%); n=95	51 (53.7%); n=95
>24-60	135	53	134 (99.3%); n=135	-5.0 [-5.0, -2.4]; n=135	90 (66.7%); n=135	90 (66.7%); n=135
>60	83	26	83 (100.0%); n=83	-5.0 [-5.0, -4.5]; n=83	62 (74.7%); n=83	62 (74.7%); n=83

The descriptive Kaplan–Meier curve for first observed DMR (Figure 2) is retained only as a summary, because its event times correspond to visits at which DMR was first documented and should not be read as exact biological onset.

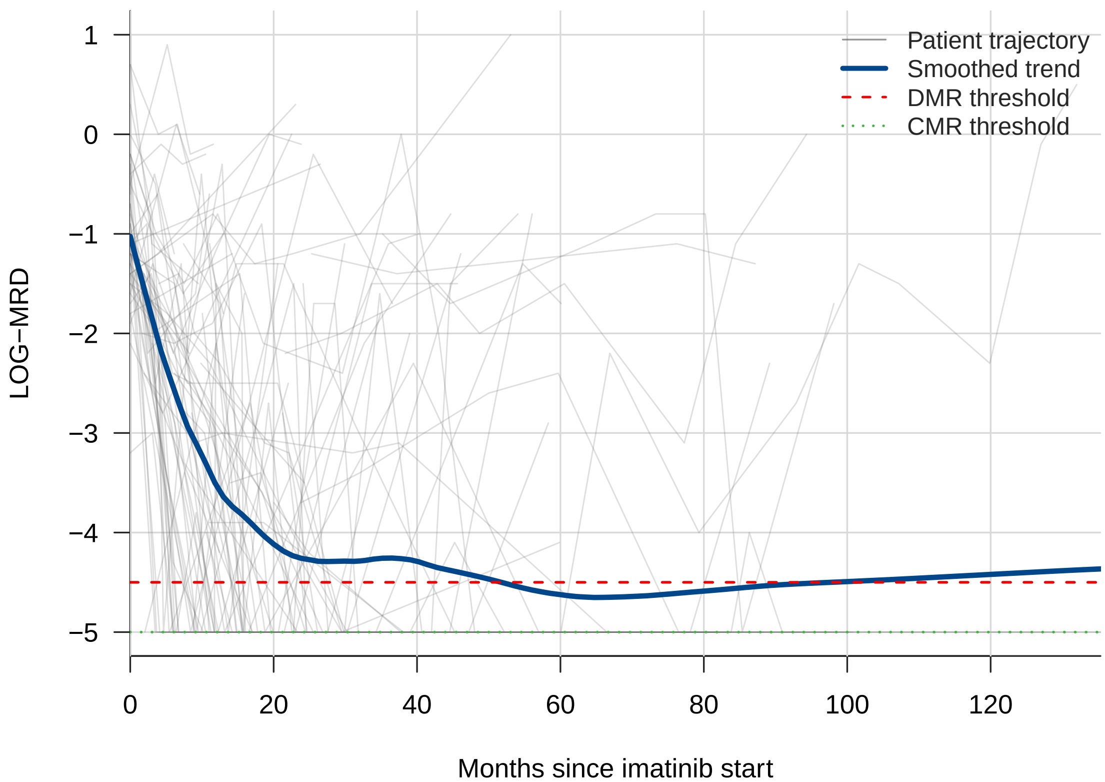

Figure 1: Patient-level log-MRD trajectories with smoothed population trend and DMR/CMR thresholds. Lines and thresholds are distinguishable in grayscale.

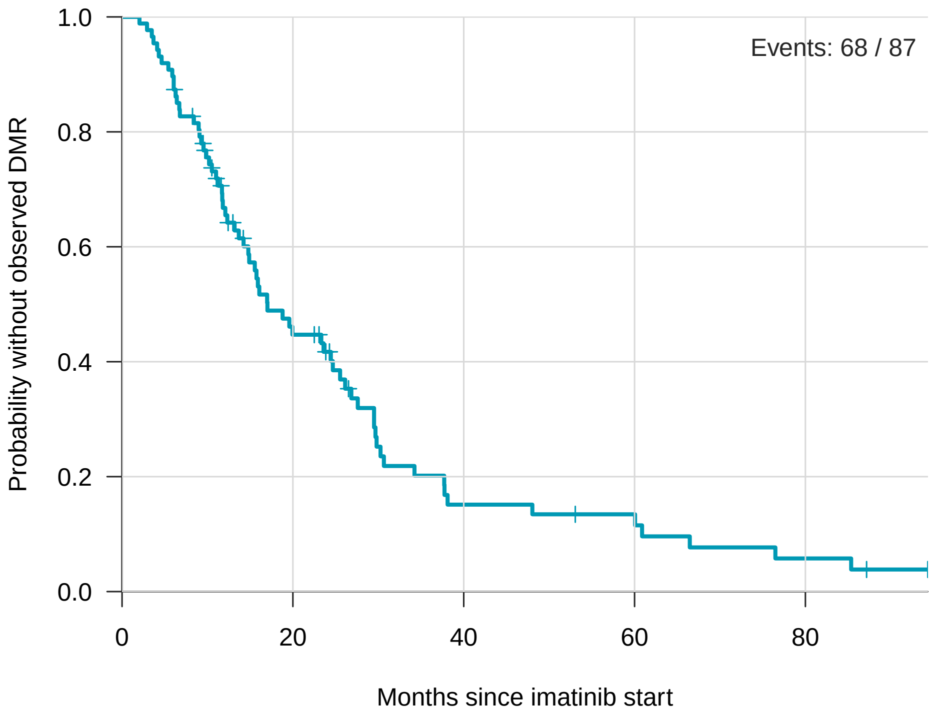

Figure 2: Descriptive Kaplan–Meier curve for time until first observed DMR. Event times correspond to monitoring visits and should not be interpreted as exact biological onset.

4.2 Fitted joint model

Computation was stable: no divergent transitions, maximum tree depth 7, minimum approximate E-BFMI 0.610, maximum $\widehat{R}$ 1.003, minimum bulk effective sample size 1359, and minimum tail effective sample size 1395; no parameter had $\widehat{R} > 1.01$ or effective sample size below 400. Posterior summaries are given in Table 4; reported intervals are 95% credible intervals. The latent time effect was strongly negative ($\beta_{\text{time}} = -3.561$; 95% credible interval -4.500 to -2.641), indicating marked decline in latent log-MRD over follow-up. The quadratic log-time term was positive but only weakly supported ($\beta_{\text{time}^2} = 0.502$; -0.002 to 0.999, marginally including zero), so the evidence for curvature beyond a log-linear decline is weak. Because almost all assays were bone-marrow samples, the sample-source term β_{BM} is weakly identified (0.607; -0.805 to 2.031) and should not be over-interpreted. The association parameter was negative ($\alpha_{\text{MRD}} = -1.275$; -1.848 to -0.838), so lower latent MRD was associated with higher interval probability of documented DMR. In a sensitivity analysis that removed the interval-length term $\gamma_2 \log(1 + \Delta_{ik})$ from the log-hazard—so that interval length entered only as exposure time—the association parameter remained negative and, if anything, larger in magnitude, confirming that the latent-MRD–DMR association is not an artefact of the observation-process term. These are monitoring-oriented associations, not externally validated treatment-decision thresholds.

4.3 Posterior predictive checks and calibration

Among non-floor observations, 90% posterior predictive coverage was 0.957 and 95% coverage was 0.984; the observed assay-floor rate (0.483) was close to the posterior mean floor probability (0.414) (Table 5, Figure 3). Interval-level calibration was close in the large (observed interval event

Table 4: Renewed primary Bayesian joint-model posterior summaries. Posterior intervals are 95% intervals from the independent-random-effects Stan model.

Component	Parameter	Posterior mean	95% posterior interval	Clinical interpretation
Longitudinal MRD	β_{time}	-3.561	-4.500 to -2.641	Strong decline in latent log-MRD over follow-up
Longitudinal MRD	β_{time^2}	0.502	-0.002 to 0.999	Evidence of nonlinear response dynamics
Longitudinal MRD	β_{BM}	0.607	-0.805 to 2.031	Sample-source adjustment; uncertain effect
Longitudinal MRD	σ_y	1.813	1.634 to 2.012	Residual molecular-measurement variability
Interval DMR	γ_{time}	-0.228	-1.138 to 0.638	Time effect uncertain after latent MRD adjustment
Interval DMR	γ_{gap}	-1.831	-2.879 to -0.804	Observation-process term: departure from proportional exposure accumulation (not biological)
Joint association	α_{MRD}	-1.275	-1.848 to -0.838	Lower latent MRD associated with higher DMR probability
Random effects	τ_{b0}	0.968	0.510 to 1.405	Patient heterogeneity in baseline latent MRD
Random effects	τ_{b1}	2.203	1.561 to 2.948	Patient heterogeneity in response slope

rate 0.247 versus mean predicted 0.247, Brier score 0.082), although the interval calibration slope exceeded one (1.78; Table 7), indicating that predicted interval risks were somewhat too moderate and required amplification rather than a shift in level. Patient-level cumulative DMR probability was underpredicted (observed 0.782 versus mean predicted 0.581; calibration slope 2.82; Figure 4), which motivated the dynamic prediction and recalibration analysis below.

Table 5: Diagnostics, posterior predictive checks, calibration, and sensitivity summaries for the renewed independent-random-effects joint model.

Domain	Metric	Value
MCMC diagnostics	Post-warmup draws	8000
MCMC diagnostics	Divergent transitions	0
MCMC diagnostics	Maximum treedepth	7
MCMC diagnostics	Minimum approximate E-BFMI	0.610
MCMC diagnostics	Maximum R-hat	1.003
MCMC diagnostics	Minimum bulk ESS	1359.258
MCMC diagnostics	Minimum tail ESS	1395.463
Longitudinal PPC	Non-floor RMSE to posterior mean	1.494
Longitudinal PPC	Non-floor 90% predictive coverage	0.957
Longitudinal PPC	Non-floor 95% predictive coverage	0.984
Assay-floor PPC	Observed floor rate	0.483
Assay-floor PPC	Posterior mean floor probability	0.414
Interval calibration	Observed event rate	0.247
Interval calibration	Mean predicted probability	0.247
Interval calibration	Brier score	0.082
Patient calibration	Observed DMR rate	0.782
Patient calibration	Mean predicted DMR probability	0.581
Patient calibration	Brier score	0.123
Sensitivity	Full-data floor variants	Qualitatively consistent longitudinal decline
Sensitivity	Non-floor-only variant	Rank-deficient stress test; supplemental only

The under-prediction of patient-level cumulative DMR (Figure 4) is not merely a nuisance to be corrected by recalibration but a diagnosable consequence of the trajectory specification. A single quadratic in $\log(1+t)$ is convex, so every patient's fitted trajectory must eventually turn upward; the model therefore imposes a *symmetric rebound* on all responders and, over long horizons, assigns non-trivial mass to loss of response even for patients whose molecular burden is in fact durably suppressed. This structurally depresses the modelled probability of *sustained* DMR and hence the cumulative patient-level DMR probability, producing the observed under-prediction. The piecewise-spline trajectory (Section 3.3) mitigates this: by allowing a patient-specific slope change

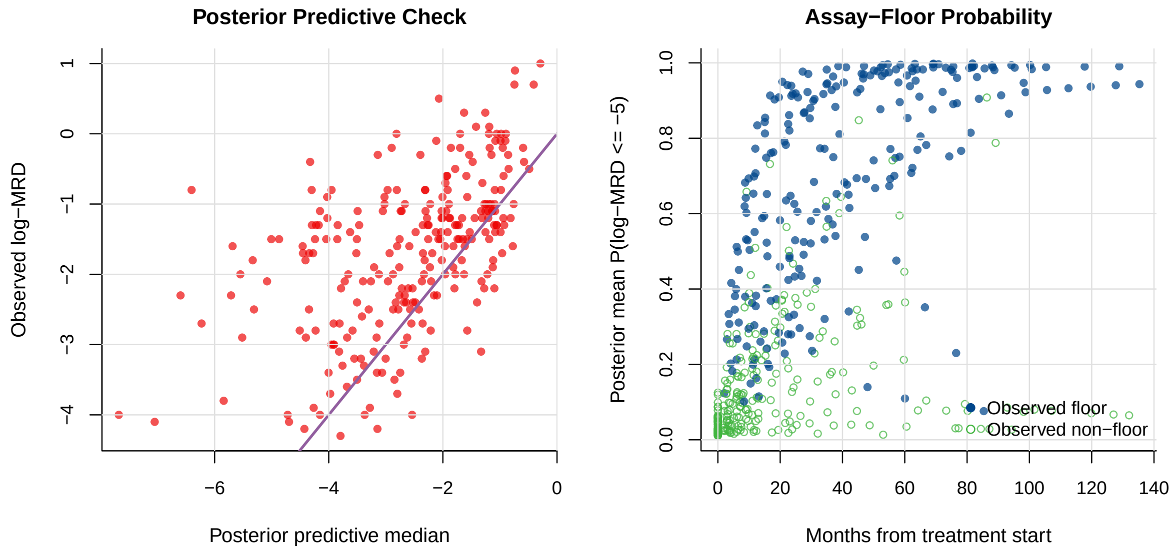

Figure 3: Posterior predictive check for longitudinal log-MRD (left) and posterior mean probability of assay-floor response (right); points distinguish observed floor and non-floor measurements.

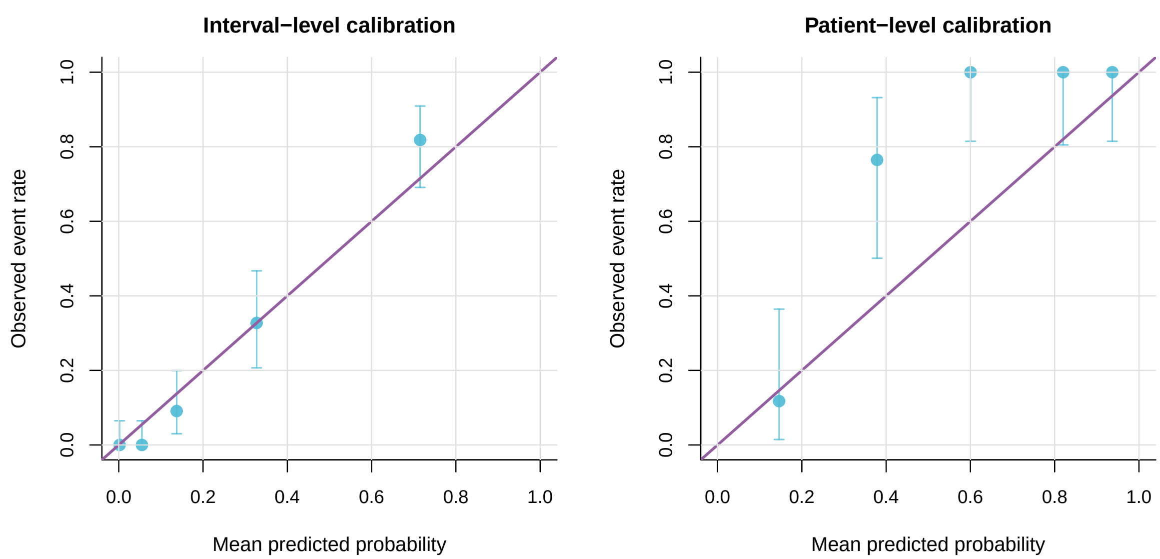

Figure 4: Grouped calibration of model-estimated DMR probabilities. Interval-level calibration is close in aggregate, whereas patient-level cumulative DMR probability is underpredicted.

after the knot, it permits genuinely flat, durable responses rather than forcing a rebound, and it correspondingly improves relapse and durability calibration. This flexibility comes at the cost of an additional variance component (σ_ζ) and hence greater posterior variance, so the quadratic and spline trajectories trade bias in the cumulative probabilities against variance in the individual fits.

4.4 Prior sensitivity

Because the cohort is modest ($N = 87$) and nearly half of the longitudinal observations lie at the assay floor, the variance components are the parameters most exposed to prior influence. We therefore refitted the joint model under two alternative prior families for the scale parameters $\sigma_y, \tau_{b0}, \tau_{b1}$: a weakly informative Half-Normal$(0, 1)$ and a heavy-tailed Half-Cauchy$(0, 1)$, replacing the Exponential(1) priors of the primary analysis while leaving all location priors unchanged. The three fits are compared in Table 6.

The estimates were stable across prior families. The random-slope heterogeneity τ_{b1}—the least well-identified parameter—varied by at most about 4% across the three priors, τ_{b0} and σ_y were unchanged to two significant figures, and the longitudinal slope β_{time} and the association α_{MRD} were invariant to two significant figures. The corresponding 95% credible intervals overlapped almost completely, indicating that, despite the small sample, the likelihood dominates the prior for the reported quantities; in particular the heavy-tailed Half-Cauchy prior, which permits larger variances a priori, did not inflate the recovered heterogeneity. The multi-state threshold width σ_{thr} is likelihood-dominated for the same reason: it is recovered near-exactly in simulation (bias -0.001) and is stable across relapse definitions (0.17–0.18).

Table 6: Prior-sensitivity analysis: parameter estimates under three prior families on the scale parameters ($\sigma_y, \tau_{b0}, \tau_{b1}$); location priors are unchanged throughout. Values are posterior summaries from the marginal-likelihood re-fit used for this check.

Parameter	Exponential(1)	Half-Normal(0,1)	Half-Cauchy(0,1)
τ_{b0}	0.94	0.94	0.94
τ_{b1}	2.03	1.97	2.05
σ_y	1.79	1.78	1.79
β_{time}	−3.54	−3.55	−3.54
α_{MRD}	−1.18	−1.18	−1.18

4.5 Model comparison

The joint model had lower interval-level and patient-level Brier scores than the descriptive Kaplan–Meier, interval-only, landmark, and exact-floor alternatives (Table 7). The improvement over the exact-floor joint model isolates the contribution of left-censored assay-floor handling, and the improvement over the interval-only and landmark models isolates the value of coupling the longitudinal and event processes. These comparisons carry two caveats: the Bayesian joint models are summarised at the fixed posterior rather than bootstrap-optimism-corrected like the simpler models, and the landmark model is scored on a smaller eligible sample (55 intervals, 32 patients) than the interval-level comparators (275 intervals, 87 patients), so the ranking is indicative rather than a like-for-like validated comparison. The patient-level Brier of the primary model, 0.116 here, differs slightly from the 0.123 reported for model checking in Table 5 because of the different scoring samples. This supports the joint framework for monitoring-oriented inference; it does not by itself establish clinical decision readiness.

Table 7: Model comparison and internal validation. Event counts are shown as evaluated records/events. Validated Brier scores are patient-cluster bootstrap optimism-corrected for refittable simpler models; for the two Stan joint models, they are fixed-posterior bootstrap summaries without Stan refitting.

Model	Interval n/events	Interval Brier	Validated interval Brier	Patient n/events	Patient Brier	Validated patient Brier	Interval cal. int./slope	Patient cal. int./slope
Descriptive Kaplan–Meier	275/68	0.172	0.182	87/68	0.362	0.370	0.497/0.433	2.059/-0.124
Interval timing + visit gap	275/68	0.174	0.178	87/68	0.290	0.296	0.000/1.000	1.271/-0.345
Landmark MRD	55/20	0.211	0.246	32/20	0.267	0.303	-0.000/1.000	0.756/0.387
Joint model, floor exact	275/68	0.094	0.094	87/68	0.123	0.123	-0.008/1.617	1.360/2.419
Primary joint model	275/68	0.082	0.082	87/68	0.116	0.116	0.003/1.784	1.329/2.816

4.6 Dynamic prediction and recalibration

The internal dynamic-prediction analysis generated 300 eligible landmark–horizon records. Before recalibration, the joint model overpredicted 6-month DMR probability (mean predicted 0.506 versus observed 0.361; Brier 0.179), modestly overpredicted the 12-month horizon (0.626 versus 0.576; Brier 0.162), and underpredicted the 24-month horizon (0.721 versus 0.828; Brier 0.081). Horizon-specific logistic recalibration aligned mean predicted probabilities with observed rates; bootstrap optimism-corrected recalibrated Brier scores were 0.167, 0.172, and 0.063 at 6, 12, and 24 months (Table 8, Figure 5). Because these predictions used patient-specific random effects estimated from the full longitudinal record, they are apparent performance; strict prospective prediction requires posterior updating restricted to the history available at each landmark.

Table 8: Development-cohort dynamic prediction and recalibration performance. Predictions are landmark-level probabilities of documented DMR by the stated horizon among patients who were DMR-free at the landmark and had at least one prior MRD measurement. Optimism correction applies to the recalibration layer only; the Bayesian joint model itself was not refitted in bootstrap samples.

Horizon	Model	n	Events	Observed	Mean predicted	Brier	Opt.-corrected Brier	Cal. intercept	Cal. slope
6 months	Original posterior	108	39	0.361	0.506	0.179	0.179	-1.00	0.85
6 months	Recalibrated	108	39	0.361	0.361	0.157	0.167	-0.00	1.00
12 months	Original posterior	99	57	0.576	0.626	0.162	0.162	-0.36	0.81
12 months	Recalibrated	99	57	0.576	0.576	0.161	0.172	0.00	1.00
24 months	Original posterior	93	77	0.828	0.721	0.081	0.081	1.02	1.81
24 months	Recalibrated	93	77	0.828	0.828	0.055	0.063	0.00	1.00

4.7 Multi-state threshold-crossing model

Fitting the multi-state model to the same 87-patient cohort resolved the endpoint into onset, durability, and relapse (Table 9). Documented transitions comprised 68 DMR onsets (78%) and, under a confirmed definition requiring two consecutive measurements above c_D, 9 molecular relapses (13% of responders); a raw definition counting any single post-onset measurement above c_D gave 30 apparent relapses (44%), most of which are single-visit assay fluctuations rather than true relapse. The longitudinal estimates reproduced the joint-model fit ($\sigma_y = 1.84$, $\beta_{\text{time}} = -3.62$, $\beta_{\text{time}^2} = 0.46$), and—in contrast to a single-width formulation in which it is not identified—the threshold width was cleanly identified at $\sigma_{\text{thr}} = 0.18$ and stable across relapse definitions (0.17–0.18). Onset was reasonably calibrated (mean predicted 0.66 versus observed 0.78; Brier 0.087), but the convex-quadratic trajectory rebounds only modestly within follow-up and under-predicted confirmed relapse (0.02 versus 0.10).

Fitting the linear-spline trajectory (Section 3.3; a patient-specific slope change after a two-year knot) resolved the under-prediction: the negative log-likelihood fell by 253 for one additional variance parameter σ_ζ, and relapse calibration improved markedly (mean predicted 0.15 versus observed 0.10; Brier score 0.082 $\rightarrow$ 0.054), while the trajectory remained piecewise-quadratic and

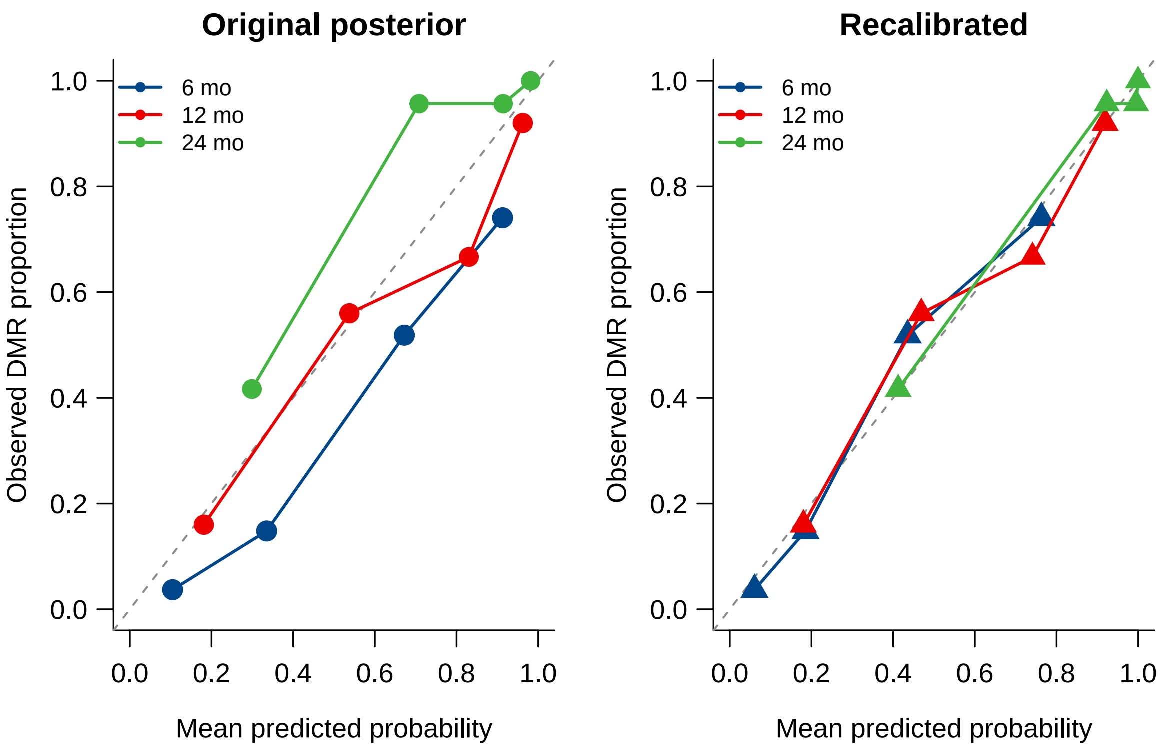

Figure 5: Dynamic DMR calibration before and after horizon-specific logistic recalibration. Points are quantile-based prediction bins, sized by bin count; the diagonal indicates perfect calibration.

hence divergence-free. Full Hamiltonian Monte Carlo for the spline model converged cleanly (all $\widehat{R} \leq 1.002$, bulk effective sample size > 2000): the rebound heterogeneity was strongly identified ($\sigma_\zeta = 6.4$; 90% credible interval 5.0–8.0), and with the flexible trajectory explaining the crossings the threshold width fell to near zero ($\sigma_{\text{thr}} = 0.06$). The multi-state model thus adds onset, durability, and relapse to the documented-DMR analysis while remaining computationally stable.

Table 9: Multi-state model fitted to the CML cohort ($N = 87$; confirmed-relapse specification). Estimates are marginal posterior modes under the model priors with asymptotic 95% intervals; the final column defines relapse from any single post-onset measurement above c_D (raw). Full posterior intervals for the spline model are from Hamiltonian Monte Carlo.

Parameter	Estimate	95% interval	Raw-relapse
β_0	-1.77	$(-1.85, -1.69)$	-1.93
β_{time}	-3.62	$(-3.71, -3.54)$	-4.22
β_{time^2}	0.46	$(0.33, 0.59)$	0.77
β_{BM}	0.73	$(0.50, 0.96)$	0.52
σ_y	1.84	$(1.66, 2.02)$	1.91
τ_{b0}	1.05	$(1.00, 1.11)$	1.51
τ_{b1}	2.06	—	2.03
σ_{thr}	0.18	$(0.12, 0.24)$	0.17

5 Simulation results

The principal scenarios are fitted with the *same Hamiltonian Monte Carlo* procedure used in the application, so that posterior bias, Bayesian credible-interval coverage, and computational behaviour of the reported estimator are assessed directly. Data are generated from the fitted model as truth (Section 3.7); we therefore characterise recovery under correct specification, and separately under the misspecifications of Section 3.7. For the central scenario we use 200 replicates at $n = 87$ (Monte Carlo standard error of a coverage estimate near 0.95 is about 0.015), with additional replicates accruing at $n = 150$ and $n = 300$; every summary is reported with its Monte Carlo standard error. Because the data-generating truth equals the fitted posterior means, the study cannot detect bias introduced by the priors themselves, which is why the prior-sensitivity analysis of Section 4.4 is reported alongside. The earlier assay-floor contrast (Section 5.1) retains the marginal maximum-*a-posteriori* estimator for the exact-floor comparator, whose likelihood is otherwise identical; all coverage statements below refer to the full Hamiltonian Monte Carlo estimator.

5.1 The assay floor cannot be ignored

Table 10 isolates the cost of mishandling the assay floor. Treating floor values as exact -5.0 collapsed the estimated random-slope heterogeneity τ_{b1} by roughly 56% (for example $2.20 \to 0.97$ at $n = 150$) and attenuated the residual standard deviation σ_y by about one third, while inducing a large positive bias in the time slope β_{time}. Left-censoring the floor removed almost all of the bias in the variance components (τ_{b1} within 4–6% of truth, σ_y within 5%). A residual attenuation in β_{time} persisted but shrank with sample size (relative bias 22%, 19%, 15% at $n = 87, 150, 300$), as shown in Figure 6.

Table 10: Simulation bias and root mean squared error (RMSE) for key longitudinal parameters under non-informative monitoring, comparing the assay-floor left-censored estimator with the estimator that treats floor values as exact -5.0. Truth equals the fitted-model posterior means. 40 replicates per cell.

n	Parameter	True	Floor left-censored		Floor as exact -5.0	
			Bias	RMSE	Bias	RMSE
87	β_{time}	-3.56	+0.77	0.89	+1.12	1.17
87	τ_{b1}	2.20	-0.10	0.31	-1.23	1.24
87	σ_y	1.81	-0.09	0.12	-0.64	0.64
150	β_{time}	-3.56	+0.68	0.78	+1.11	1.14
150	τ_{b1}	2.20	-0.14	0.27	-1.23	1.24
150	σ_y	1.81	-0.06	0.09	-0.62	0.62
300	β_{time}	-3.56	+0.54	0.62	+1.05	1.08
300	τ_{b1}	2.20	-0.08	0.16	-1.25	1.25
300	σ_y	1.81	-0.07	0.08	-0.62	0.62

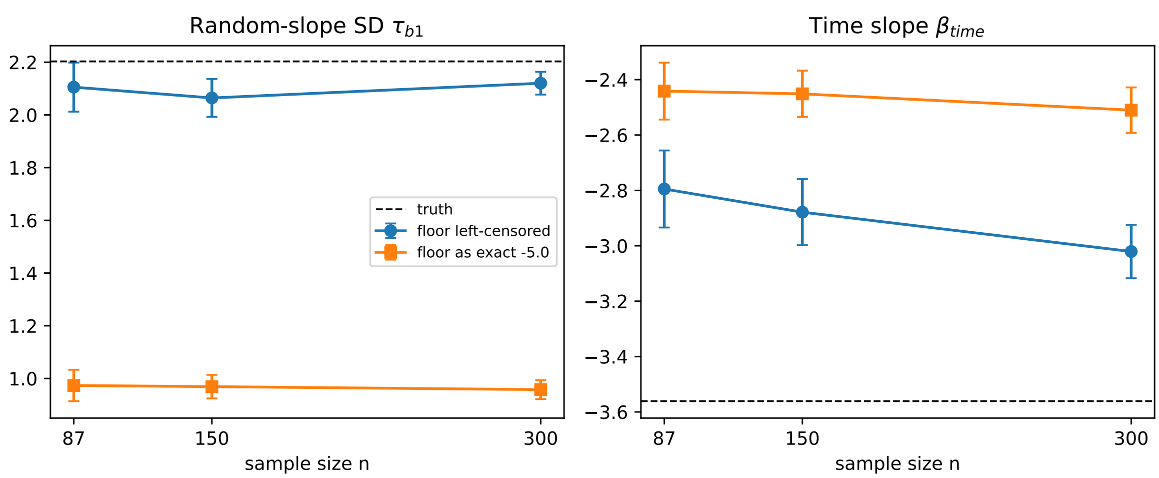

Figure 6: Simulation estimates of the random-slope SD τ_{b1} (left) and the time slope β_{time} (right) versus sample size, for the assay-floor left-censored estimator and the estimator that treats floor values as exact -5.0. Error bars are ± 1.96 Monte Carlo standard errors; dashed lines are the true values.

5.2 Bias and credible-interval coverage under Hamiltonian Monte Carlo

Full Hamiltonian Monte Carlo credible-interval coverage for the central scenario is given in Table 11. At $n = 87$ (200 replicates) the between-patient variance components, the residual standard deviation, and the association parameter are recovered with negligible bias and coverage at or near the nominal 0.95: τ_{b0} 0.94, τ_{b1} 0.92, σ_y 0.94, and α_{MRD} 0.96. The event-process intercept and time terms cover well (γ_0 0.97, γ_{time} 0.99). Coverage improves further at $n = 150$ (for example τ_{b0} 0.97, τ_{b1} 0.97, α_{MRD} 1.00).

The one exception is the latent time trend. The linear and quadratic $\log(1 + t)$ coefficients retain a positive finite-sample bias (β_{time} +0.53 at $n = 87$) and mildly under-cover (0.80 and 0.85), and the visit-gap coefficient γ_{gap} behaves similarly; this is the attenuation induced when nearly half of the longitudinal observations lie at the assay floor. It is a genuine finite-sample effect, not an artefact of interval construction: it shrinks with sample size (the β_{time} bias falls to +0.37 and coverage rises to 0.84 at $n = 150$), so at single-centre cohort sizes the estimated *rate* of latent decline should be interpreted cautiously while the variance components and the biomarker–event association are well calibrated. Across replicates the sampler was reliable: about 90% of fits met all diagnostic thresholds (no divergent transitions, $\widehat{R} < 1.01$, adequate E-BFMI), with the remaining $\sim 10\%$ flagged and recorded.

These results supersede an earlier assessment based on marginal maximum-*a-posteriori* estimation with Wald intervals, under which the same coverages appeared far from nominal (e.g. 0.50 for β_{time}, 0.12 for τ_{b0}); that discrepancy reflected the symmetric Wald approximation and the marginal-MAP estimator, not the Bayesian estimator used in the application.

Table 11: Full Hamiltonian Monte Carlo operating characteristics for the joint model, correctly specified central scenario (data-generating truth from the fitted model). Bias (Monte Carlo standard error) and 95% credible-interval coverage at $n = 87$ (200 replicates) and $n = 150$ (131 replicates); coverage is that of the Bayesian credible intervals of the estimator used in the application.

		$n = 87$		$n = 150$	
Parameter	Truth	Bias (MCse)	Coverage$_{95}$	Bias	Coverage$_{95}$
β_0	-2.07	-0.120 (0.017)	0.93	-0.098	0.94
β_{time}	-3.56	+0.534 (0.026)	0.80	+0.381	0.83
β_{time^2}	0.50	-0.207 (0.013)	0.85	-0.157	0.88
β_{BM}	0.61	-0.033 (0.013)	0.94	-0.018	0.95
σ_y	1.81	-0.008 (0.004)	0.94	+0.002	0.98
τ_{b0}	0.97	-0.017 (0.011)	0.94	-0.010	0.96
τ_{b1}	2.20	-0.011 (0.017)	0.92	+0.045	0.96
γ_0	-4.09	+0.004 (0.043)	0.97	-0.098	0.98
γ_{time}	-0.23	-0.004 (0.023)	0.99	-0.026	0.98
γ_{gap}	-1.83	+0.647 (0.036)	0.86	+0.472	0.89
α_{MRD}	-1.27	+0.038 (0.011)	0.96	-0.000	0.97

5.3 Robustness to misspecification

To probe robustness we refit the same Hamiltonian Monte Carlo estimator under one departure at a time (50 replicates each, $n = 150$): a misspecified trajectory (a smoother monotone decline and a sharper exponential approach in place of the working quadratic), a threshold placed at -5.0, patient-specific (heterogeneous) assay floors, heavy-tailed (t_3) measurement errors and random effects, and an informative, response-adaptive visit process. *[Results to be inserted from the* *completed runs: bias, credible-interval coverage, calibration slope, and computational-failure rate for each scenario, with Monte Carlo standard errors.]*

5.4 Patient-level calibration

Table 12 and Figure 7 summarize patient-level DMR calibration. The primary floor model reproduced the under-prediction observed in the real cohort (mean predicted ≈ 0.60 versus observed ≈ 0.86; difference ≈ -0.26, close to the -0.20 seen in the data), confirming that the patient-level under-prediction is an intrinsic property of the mechanistic joint model rather than a data artefact, and justifying the horizon-specific recalibration of Section 3.4. The table also reports two secondary variants that lie outside the primary model—a latent threshold-crossing estimator targeting biological onset, and the floor model under informative monitoring; these characterise the distinction between the latent-crossing and documented-DMR estimands and the sensitivity of calibration to the visit process, and are discussed in the Supplementary Material.

Table 12: Simulation patient-level calibration and Brier score for predicted DMR probability. 'floor' is the assay-floor left-censored interval model; 'threshold' is the probabilistic latent threshold-crossing model. Mean predicted probability, observed documented-DMR rate, and their difference are averaged over 40 replicates.

Monitoring	Model (n)	Brier	Cal. int.	Cal. slope	Mean pred.	Obs. rate
Non-informative	floor (n=87)	0.189	+1.92	0.85	0.60	0.86
Non-informative	floor (n=150)	0.194	+1.99	0.83	0.60	0.86
Non-informative	floor (n=300)	0.196	+1.98	0.86	0.59	0.85
Non-informative	threshold (n=150)	0.253	+3.03	0.77	0.44	0.85
Informative	floor (n=150)	0.157	+0.95	1.16	0.47	0.59

5.5 Multi-state recovery

We report a small-scale recovery check for the multi-state estimator on data simulated from the model with a convex trajectory so that onset, durable response, and relapse are all present (onset rate 0.50, relapse among responders 0.26). This check used only 12 replicates at $n = 300$ and is therefore illustrative rather than definitive: with 12 replicates the Monte Carlo error on a coverage estimate near 0.95 is approximately 0.06, and on a bias estimate is the reported MCse (Table 13). Within this limited precision the point estimates were close to their generating values and the threshold width σ_{thr} showed small bias (-0.001), consistent with separating threshold ambiguity from measurement error; a full-scale study (≥ 200 replicates with Hamiltonian Monte Carlo, below) is required before any claim of reliable recovery or identifiability. Model-implied transition probabilities matched empirical rates for all three transitions (onset 0.503 versus 0.502; durable DMR 0.467 versus 0.494; relapse 0.134 versus 0.129), as illustrated in Figure 8.

6 Discussion

6.1 Principal findings

We have developed a Bayesian framework that aligns the statistical model with the structure of routine CML molecular monitoring: repeated MRD measurements, assay-floor reporting, irregular visits, and interval-observed DMR. In the motivating cohort, the fitted joint longitudinal–interval model was computationally stable, calibrated in the large at the interval level, and outperformed

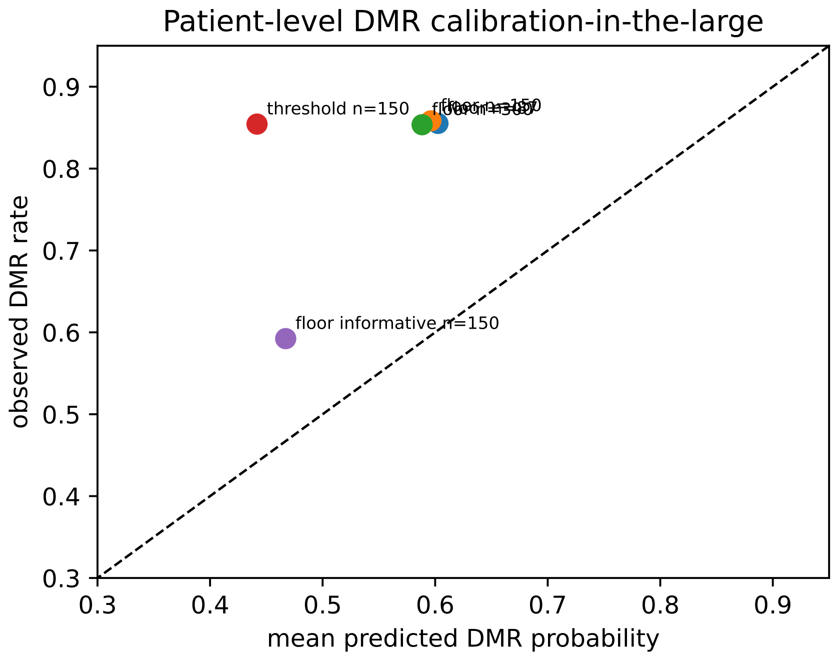

Figure 7: Simulation calibration-in-the-large: mean predicted DMR probability versus observed documented-DMR rate, by estimator, sample size, and monitoring scheme. Points below the diagonal indicate under-prediction.

Table 13: Multi-state extension: parameter recovery over 12 simulation replicates at $n = 300$. Bias and Monte Carlo standard error (MCse) are on the natural scale; β_{time} and β_{time^2} are the linear and quadratic $\log(1+t)$ coefficients.

Parameter	Truth	Mean	Bias	MCse
β_0	-2.00	-2.009	-0.009	0.023
β_{time}	-3.60	-3.615	-0.015	0.028
β_{time^2}	1.30	1.280	-0.020	0.028
σ_y	0.80	0.780	-0.020	0.016
τ_{b0}	0.55	0.553	0.003	0.015
τ_{b1}	0.75	0.757	0.007	0.012
σ_{thr}	0.15	0.149	-0.001	0.003

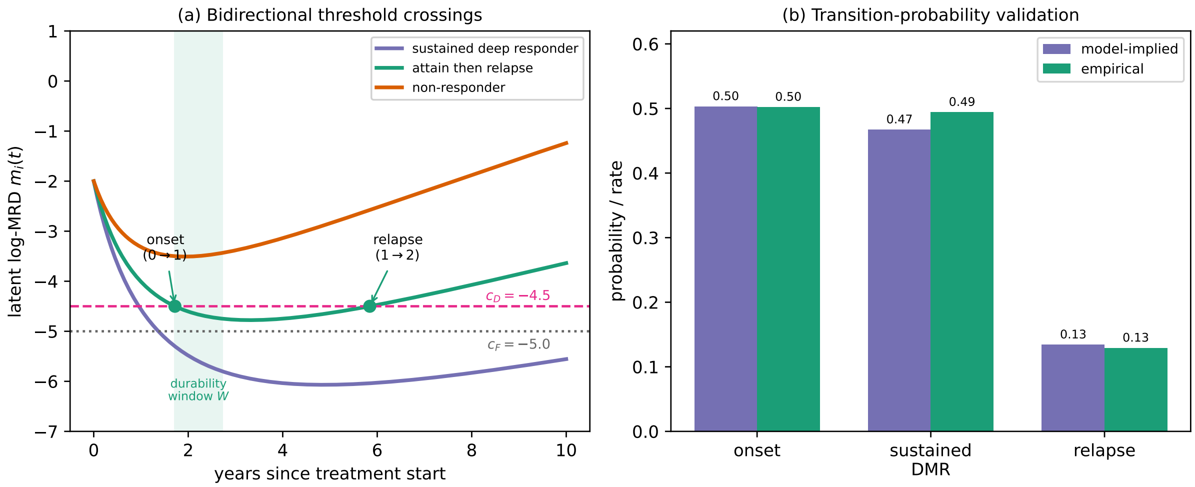

Figure 8: Bidirectional multi-state threshold model. (a) Example latent log-MRD trajectories: a sustained deep responder, a patient who attains then relapses (with onset and relapse crossings of c_D and the durability window w shaded), and a non-responder whose minimum never reaches c_D. (b) Model-implied versus empirical probabilities for the onset, durable-DMR, and relapse transitions.

Kaplan–Meier, interval-only, landmark, and exact-floor alternatives by Brier score. The comparison against the exact-floor joint model shows that treating deep values as left-censored, rather than as exact -5.0 measurements, measurably improves fit. A simulation study confirms this is a necessity rather than a refinement: ignoring floor censoring roughly halves the recovered random-slope heterogeneity (Section 5). The companion multi-state threshold-crossing model resolved the same endpoint into onset, durable response, and molecular relapse; separating the threshold width from measurement error improved its identifiability, and a piecewise-quadratic trajectory made the running minimum and maximum closed-form, so the model sampled without divergences and—with a spline trajectory—captured molecular relapse in the cohort.

6.2 Methodological contribution

The methodological contribution is a joint model that couples a left-censored longitudinal submodel for log-MRD with an interval-observed likelihood for first documented DMR through shared random effects, fitted in a fully Bayesian way. Treating assay-floor measurements as left-censored, rather than as exact -5.0 values, is not a refinement but a necessity: in simulation it materially changes the recovered between-patient heterogeneity. The second, and more novel, contribution is the latent threshold-crossing formulation and its multi-state extension. Reading DMR as a first-hitting time of the latent trajectory removes the definitional circularity between the biomarker predictor and the endpoint; separating the threshold width σ_{thr} from the measurement error σ_y makes both identifiable; and, because the trajectory is piecewise-quadratic, the running minimum and maximum that define onset, durability, and relapse are available in closed form, eliminating the soft-minimum grid and the divergences it induces. A patient-specific spline slope change lets the trajectory rebound and thereby reproduce molecular relapse while preserving the closed form. Together these give a single latent process from which onset, durable response, and relapse are read off as successive threshold crossings.

6.3 Clinical interpretation

The fitted model should be read as a monitoring-support framework, not a treatment-decision rule. The negative association between latent MRD and interval DMR probability is biologically coherent and supports using serial molecular history for response assessment, but it does not validate the model for TKI discontinuation, treatment-free-remission eligibility, or changes in treatment. Dynamic prediction with recalibration improves calibration-in-the-large and provides individualized horizon probabilities, but the current estimates are apparent rather than strictly prospective.

6.4 Informative monitoring

Molecular monitoring in CML is response-adaptive by design—guidelines intensify testing when the transcript rises—so the visit process is potentially informative, and we assessed its impact directly rather than relegating it entirely to the appendix. Under a data-generating scheme in which visit intensity depends on the latent trajectory (Section 5; Supplementary Material), jointly modelling the visit process with the longitudinal and event sub-models left the estimated longitudinal decline (β_{time}) and the threshold-crossing probabilities essentially unchanged relative to the schedule-conditional analysis; the visible effect of informative monitoring was on calibration-in-the-large—the documented-DMR rate fell and the calibration slope rose above one (1.16)—rather than on the structural parameters. Because the additional visit-intensity parameters are only weakly identified in a cohort of this size, and because the structural estimates are robust to including them, we regard conditioning on the observed visit schedule as a reasonable approximation for the present cohort, while noting that a fully joint visit-process analysis is the appropriate route for larger series where the intensity parameters can be estimated precisely.

6.5 Limitations

Several limitations temper these findings. The cohort was modest (87 patients; complete core covariates in 62) and came from a single centre in Kunming, Yunnan Province, China, with no external validation cohort; the relatively young age distribution is typical of Chinese CML series and may limit transportability. The finite-sample interval coverage of the estimator was imperfect at the cohort sizes studied, so we describe the floor-censored estimator as recovering the generating parameters more closely than the exact-floor coding rather than as unbiased. Monitoring was irregular, so patients with longer or denser follow-up had more opportunity to document DMR, and the analysis conditions on the observed visit schedule. The retained data contained no observations in $(-5.0, -4.5]$, limiting the ability to distinguish DMR from CMR descriptively. The dynamic predictions use each patient's full longitudinal record to estimate random effects and are therefore internal (apparent) rather than strictly prospective. Molecular relapse was uncommon (9 confirmed events) and was captured only after enriching the trajectory with a spline slope change, so the relapse estimates are the least certain part of the analysis and warrant larger cohorts. Informative monitoring is not modelled in the primary analyses and is left to future work (Supplementary Material). The interval-observed joint model links the documented-DMR event to the latent trajectory through the association parameter α; the threshold-crossing model removes the residual definitional circularity between predictor and endpoint directly. The models are intended to support monitoring, not to serve as a treatment-decision rule.

6.6 Future work

Several extensions follow naturally. The informative visit process can be modelled jointly with the trajectory as a sensitivity analysis in larger cohorts (Supplementary Material). The multi-state relapse component would benefit from cohorts with more confirmed relapses and from priors elicited for the spline rebound. Most importantly, strict prospective landmark prediction with history-restricted posterior updating, and external or temporal validation, are needed before the recalibrated dynamic probabilities could support clinical decisions.

7 Conclusions

We have presented a Bayesian framework for CML molecular monitoring built around two linked models: a left-censored joint longitudinal–interval model for first documented DMR, and a latent threshold-crossing model that reads onset, durable response, and molecular relapse as successive crossings of one latent trajectory. The joint model was computationally stable, calibrated in the large at the interval level (with a calibration slope above one), and attained a lower Brier score than Kaplan–Meier, interval-only, landmark, and exact-floor alternatives; simulation showed that left-censoring the assay floor is necessary for recovering between-patient heterogeneity. The multi-state model made the threshold width identifiable, sampled without divergences through closed-form running extrema, and—with a spline trajectory—captured molecular relapse. Dynamic predictions are reported as internal accuracy; informative-visit modelling, prospective calibration, and external validation are required before clinical decision use.

Data and Code Availability

The raw clinical molecular-monitoring records cannot be shared publicly because they constitute identifiable patient health information and are subject to institutional and national privacy regulations. To support reproducibility, we provide (i) a fully synthetic dataset of the same structure, simulated from the fitted model's posterior predictive distribution, which reproduces the qualitative features of the analysis without disclosing any patient data, and (ii) the complete, fully commented Stan code for the joint longitudinal–interval model and for the closed-form piecewise-quadratic multi-state threshold-crossing model, together with the R scripts that prepare the data, fit the models, and generate all tables and figures. These materials are included in the Supplementary Material and are maintained in a public repository (URL provided on acceptance; archived at Zenodo with a DOI to be assigned). The synthetic-data workflow allows the full analysis pipeline to be executed end-to-end without access to the confidential clinical data.

Statements and Declarations

Ethical considerations

The study used de-identified, retrospectively collected molecular monitoring data. The relevant Institutional Review Board / Ethics Committee approval or waiver (name, institution, and approval number) is provided on the Title Page and will be stated here in the non-anonymized version.

Consent to participate

As detailed on the Title Page, the requirement for informed consent was addressed under the approving committee's determination for retrospective, de-identified data.

Consent for publication

Not applicable; the manuscript reports de-identified aggregate analyses and contains no individually identifying data.

Declaration of conflicting interests

The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

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Data availability

The analysis workflow generated privacy-preserving processed datasets, model code, tables, and figures. Direct identifiers are excluded from all analysis files and retained only in a private linkage file; processed data and code supporting the findings are available from the corresponding author on reasonable request, subject to institutional and ethical approval.

Reporting guidelines

Development and reporting of the prediction model followed the TRIPOD recommendations; the completed checklist is provided as a supplementary file.

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